

Kris(Qi) Zhou

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Experiences

Industrial Securities Co., Ltd.

Mar. 2025 – Sep. 2025

Software Engineer Intern, Developer Tools & Experimentation Platform

Beijing, China

- Built internal **Java developer tooling for AST-based static analysis** with JavaParser across a **200K+ LOC codebase spanning 50+ services**, automating repeatable validation and reducing manual code inspection.
- Redesigned **PostgreSQL dependency-graph traversal algorithms**, cutting analysis time by **2×** and reducing peak memory usage by **40%**.
- Built real-time monitoring infrastructure for **3K+ concurrent experiments using Kafka and Flink**, processing rollout signals with **sub-second latency**.
- Partnered with engineers to improve **failure diagnostics, dependency-impact analysis, validation rules, and monitoring workflows** used before production deployment.

Guotai Haitong Securities

Jan. 2024 – May 2024

Full-Stack Software Engineer Intern, Market Data Platform

Beijing, China

- Built distributed backend services for a real-time market-data platform, ingesting **Level 2 streaming data through Kafka** and exposing services with **Spring Boot and Spring Cloud**.
- Automated testing, container image builds, and **Kubernetes deployments** through a self-hosted GitLab CI/CD pipeline.
- Developed **JUnit 5 and JMeter regression/performance tests** for core services and high-throughput workflows.
- Documented **40+ REST APIs** with Swagger/OpenAPI and developed operational interfaces for real-time monitoring workflows.

Institute of Economics, Chinese Academy of Social Sciences

Mar. 2023 – Aug. 2023

Research Assistant, JVM Testing Systems

Beijing, China

- Designed a **Monte Carlo** mutation strategy **in Java** over JVM bytecode execution paths, uncovering defects in production JVM implementations. Developed a coverage guided evolutionary algorithm that steered mutants toward higher branch, code and JVM instruction coverage.
- Fuzzed **HotSpot and OpenJ9** and diffed their execution behavior, surfacing **50+ defects including 3 critical issues** reported upstream to the **IBM OpenJ9** team.
- Organized defect cases with triggering inputs, observed behavior, coverage signals, and reproduction notes for upstream review.

Projects

CodeGraph Copilot — Developer Infrastructure & Service Dependency Intelligence

- Built a **Java/Python developer infrastructure platform** that parsed source code, modeled cross-service dependencies, and generated automated validation reports for engineering review.
- Implemented **AST-based static analysis and graph traversal algorithms** to detect risky dependency changes, missing instrumentation, unused APIs, inconsistent patterns, and valid.
- Designed PostgreSQL schemas for **files, symbols, call edges, dependency paths, build records, validation rules, and issue fingerprints**, enabling structured service-impact and failure analysis.
- Integrated **AI-assisted diagnostics** that transformed deterministic analysis results into root-cause summaries, dependency-impact explanations, recurring failure patterns, test suggestions, and CI/CD debugging guidance.

Skills

Languages: Python, TypeScript, JavaScript, Java, Go, C/C++, SQL, Bash/Shell

Frameworks: React.js, Next.js, Node.js, Express.js, FastAPI, Django, Spring Cloud, Vue.js

AI / ML: PyTorch, Hugging Face, LLMs, Transformers, LoRA/QLoRA, Model Evaluation, Deep Learning

Cloud & Infrastructure: AWS (EKS, EC2, S3), Google Cloud, Docker, Kubernetes, Linux/Unix, GitLab CI/CD, Git

Data & Developer Tools: PostgreSQL, MongoDB, Redis, Elasticsearch, Kafka, Flink, JavaParser, JUnit 5

Education

University of Washington

Sept. 2025 – June 2027

M.S. in Information Management (Specialization in Data Science & AI) (GPA: 3.9/4.0)

Seattle, WA

Peking University

Sept. 2023 – June 2025

Minor in Economics (GPA: 3.8/4.0)

Beijing, China

Beijing International Studies University

Sept. 2021 – June 2025

B.A. in Turkish (GPA: 3.9/4.0)

Beijing, China